Exploratory Data Analysis Late Nights at MOMA PS1

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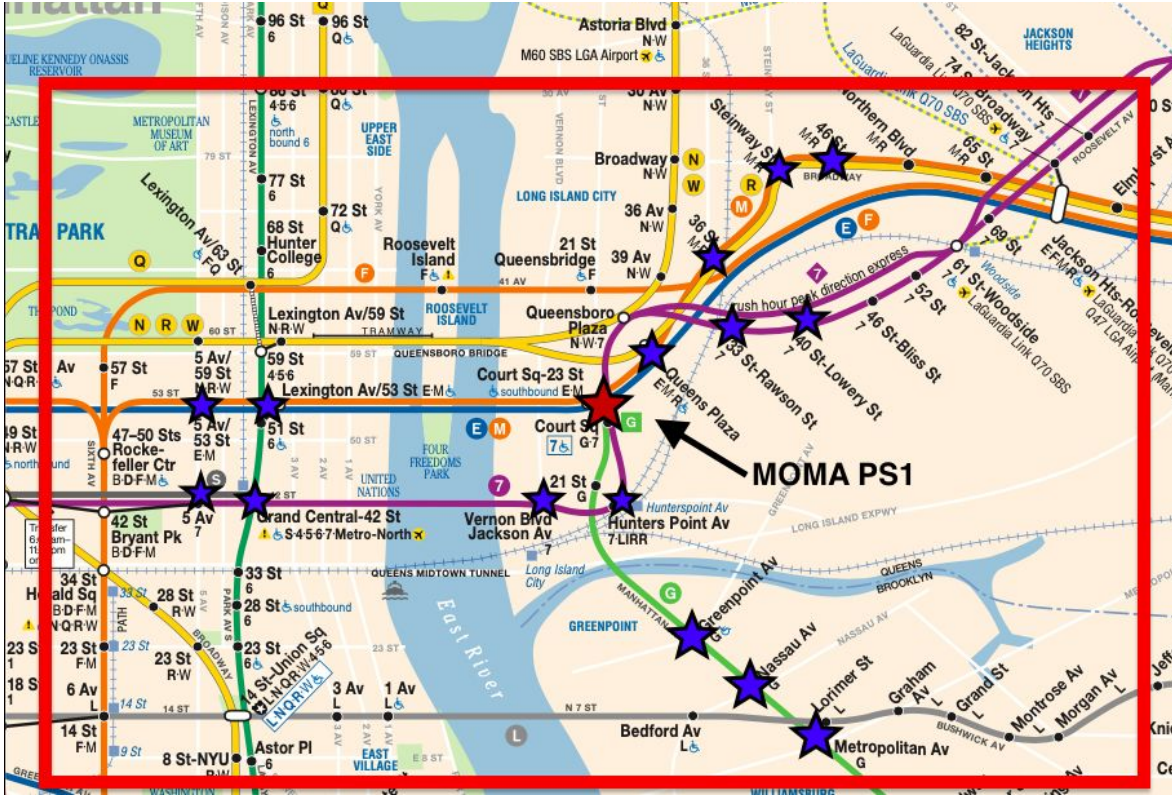
The Client

MOMA PS1

When should the museum schedule “Late Nights MOMA PS1,” an initiative to boost weekday visitor metrics through extending hours and discounting admission?

Model

Evaluate weekly and daily traffic patterns at stations with direct service to Court Square, within three stops of Court Square.



Lines of Interest

The E

Financial District to Jamaica Plain, Queens

The M

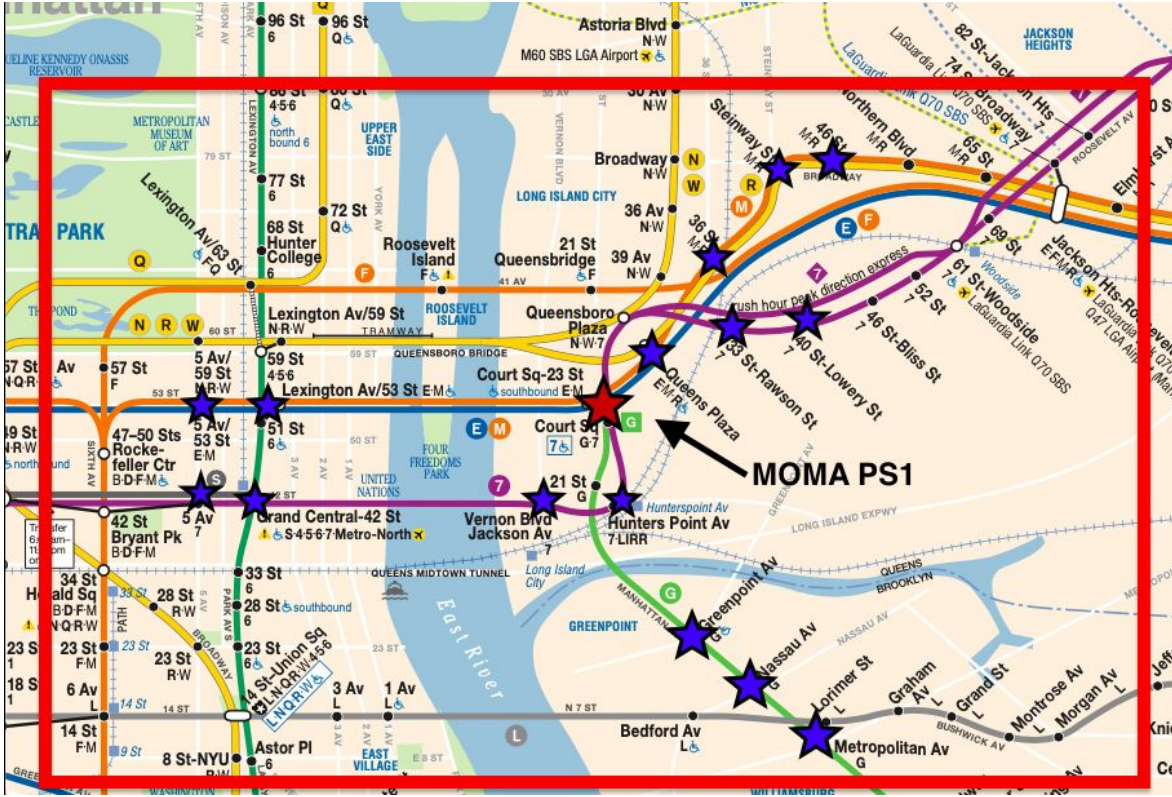
Queens to Manhattan to Brooklyn

The 7

Times Square to Flushing, Queens

The G

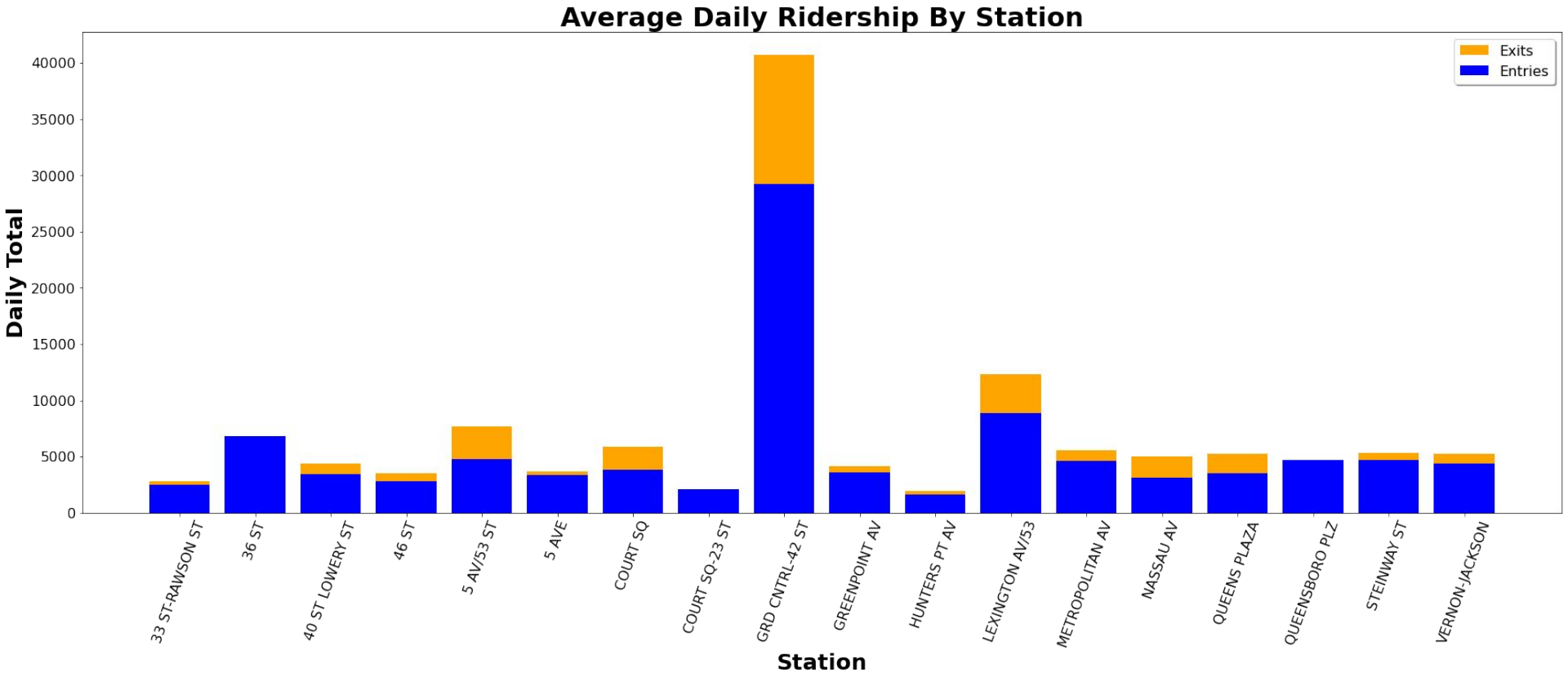
Brooklyn to Queens

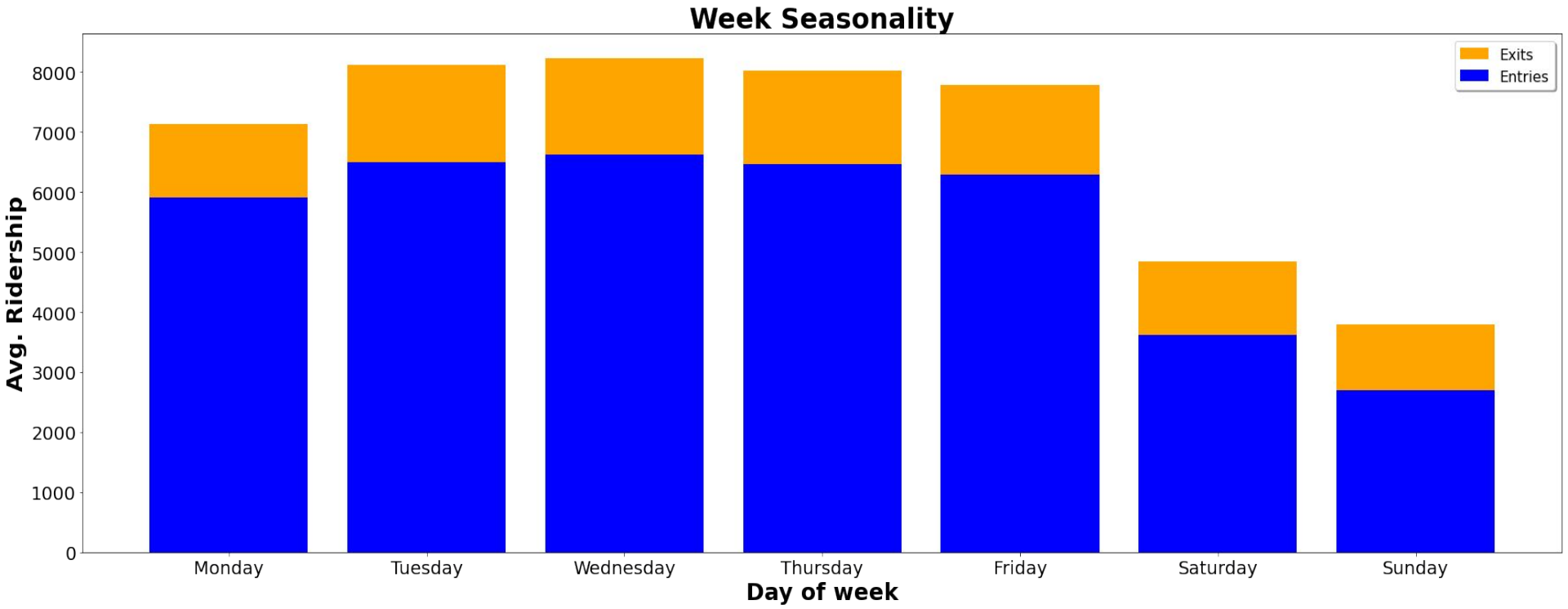


The Data & Tools

- MTA Turnstile Data at the Station Level
  - 2021 Calendar Year
  - 120,235 Observations.
- Pandas, SQLalchemy for data manipulation
- Matplotlib, and Seaborn for plotting

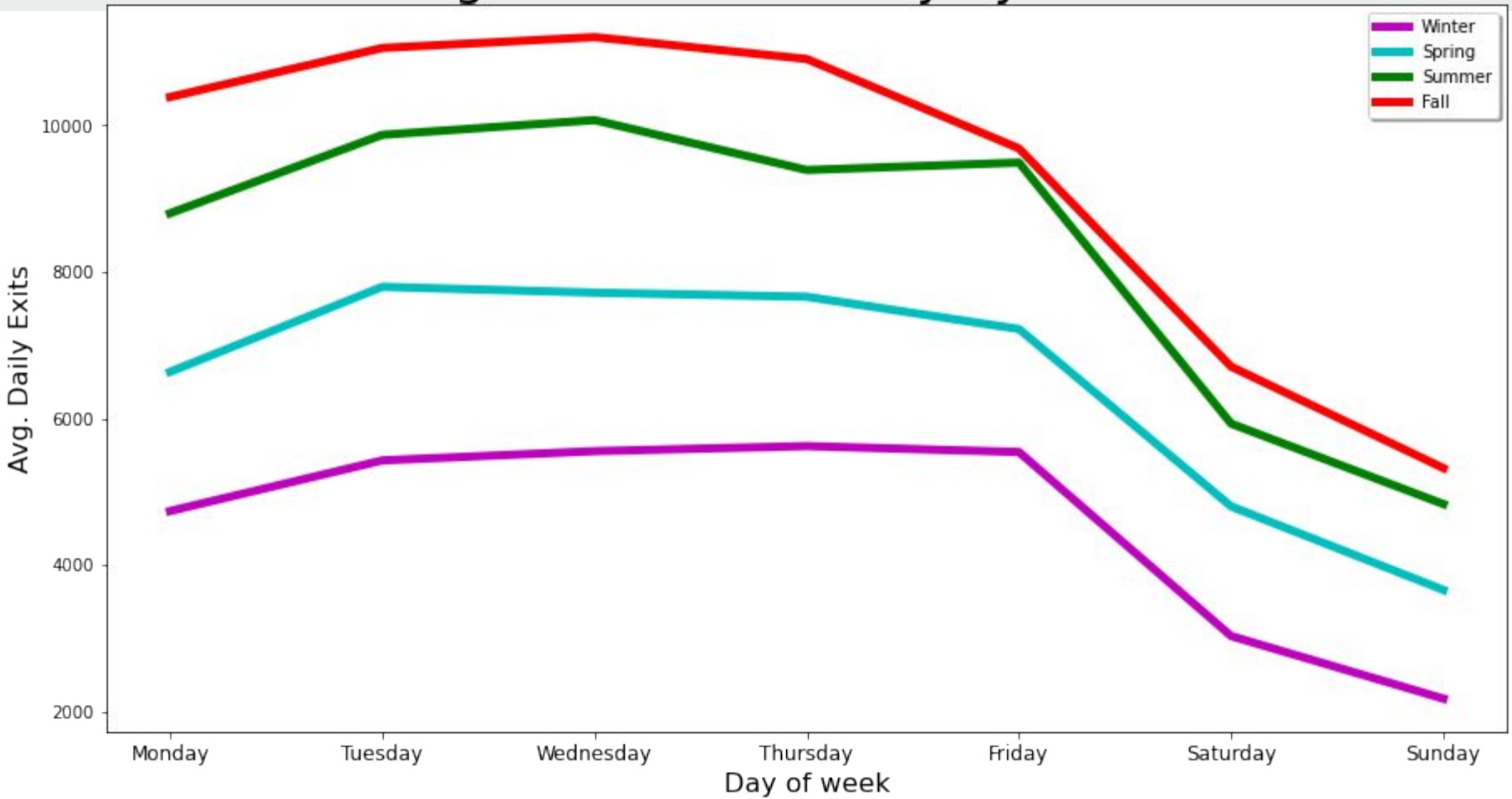
Determining the Date


Average Daily Ridership By Station
Exits
Entries
Daily Total
40000
35000
30000
25000
20000
15000
10000
5000
0
33 ST-RAWSON ST
36 ST
40 ST LOWERY ST
46 ST
5 AV/53 ST
5 AVE
COURT SQ
COURT SQ-23 ST
GRD CNTRL-42 ST
GREENPOINT AV
HUNTERS PT AV
LEXINGTON AV/53
METROPOLITAN AV
NASSAU AV
QUEENS PLAZA
QUEENSBORO PLZ
STEINWAY ST
VERNON-JACKSON
Station

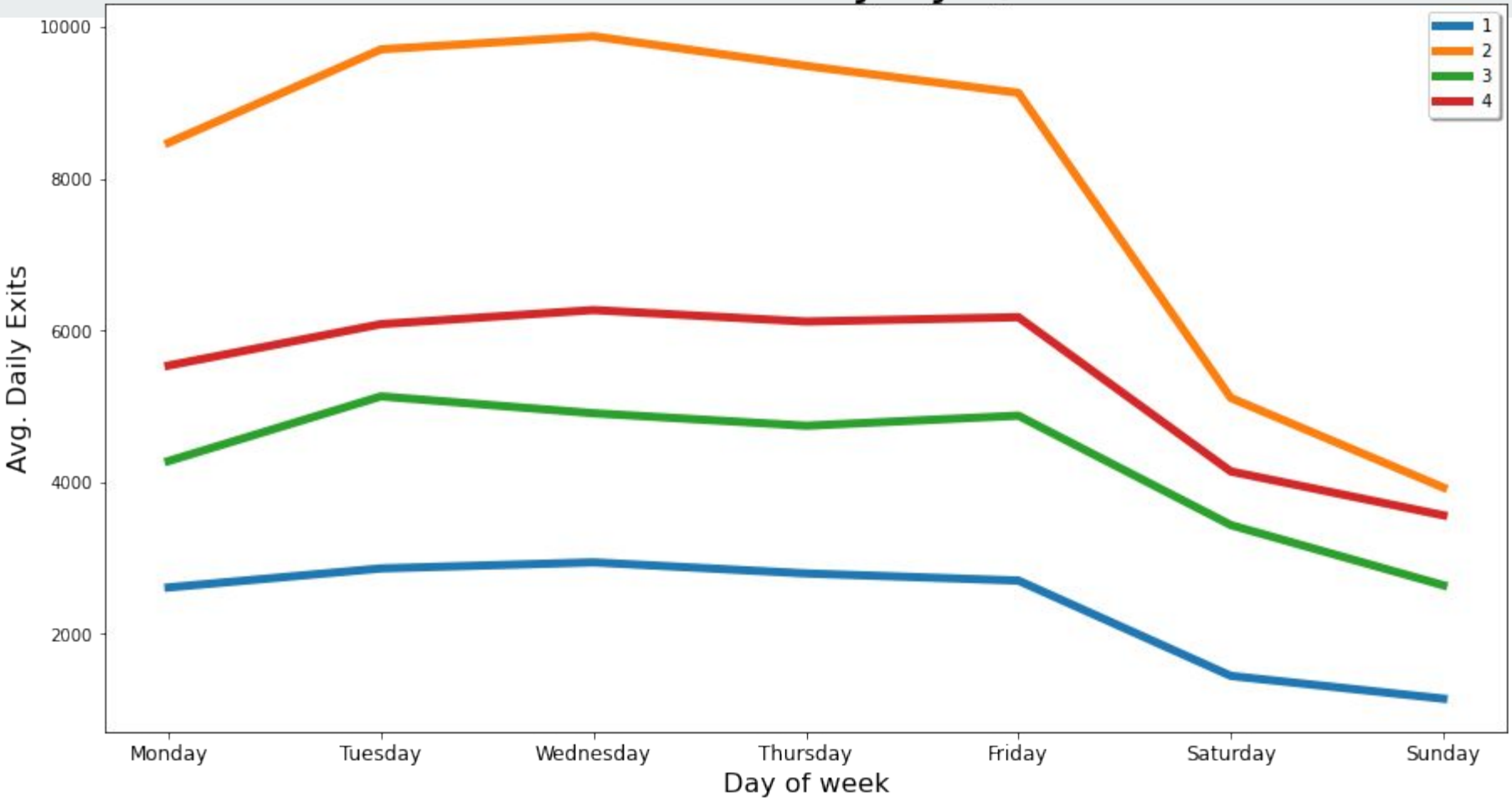

Week Seasonality
Exits
Entries
Avg. Ridership
0
1000
2000
3000
4000
5000
6000
7000
8000
Monday
Tuesday
Wednesday
Thursday
Friday
Saturday
Sunday
Day of week

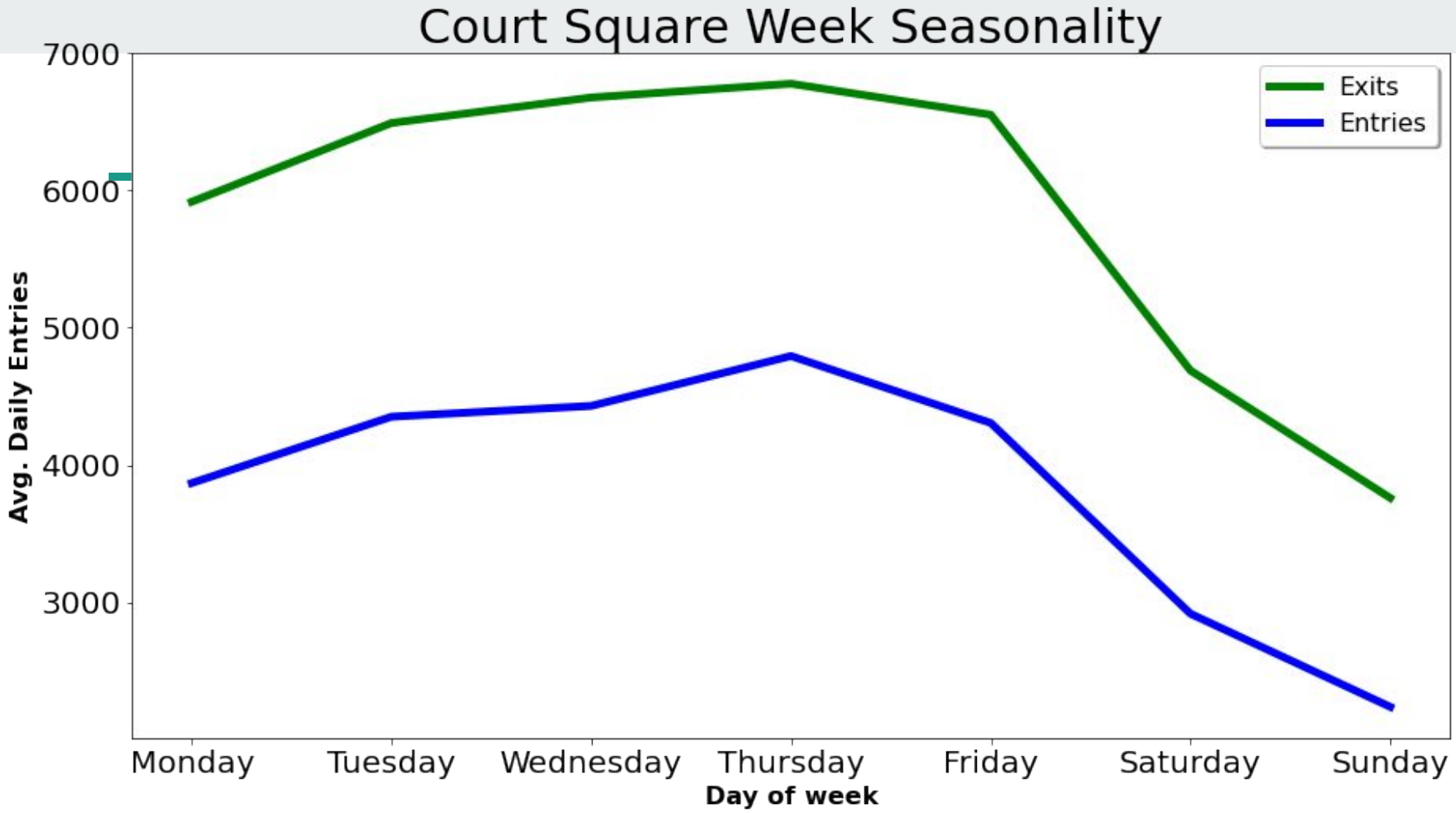
Sanity Checks: Weather & Quartile

Avg Week Seasonality by Season



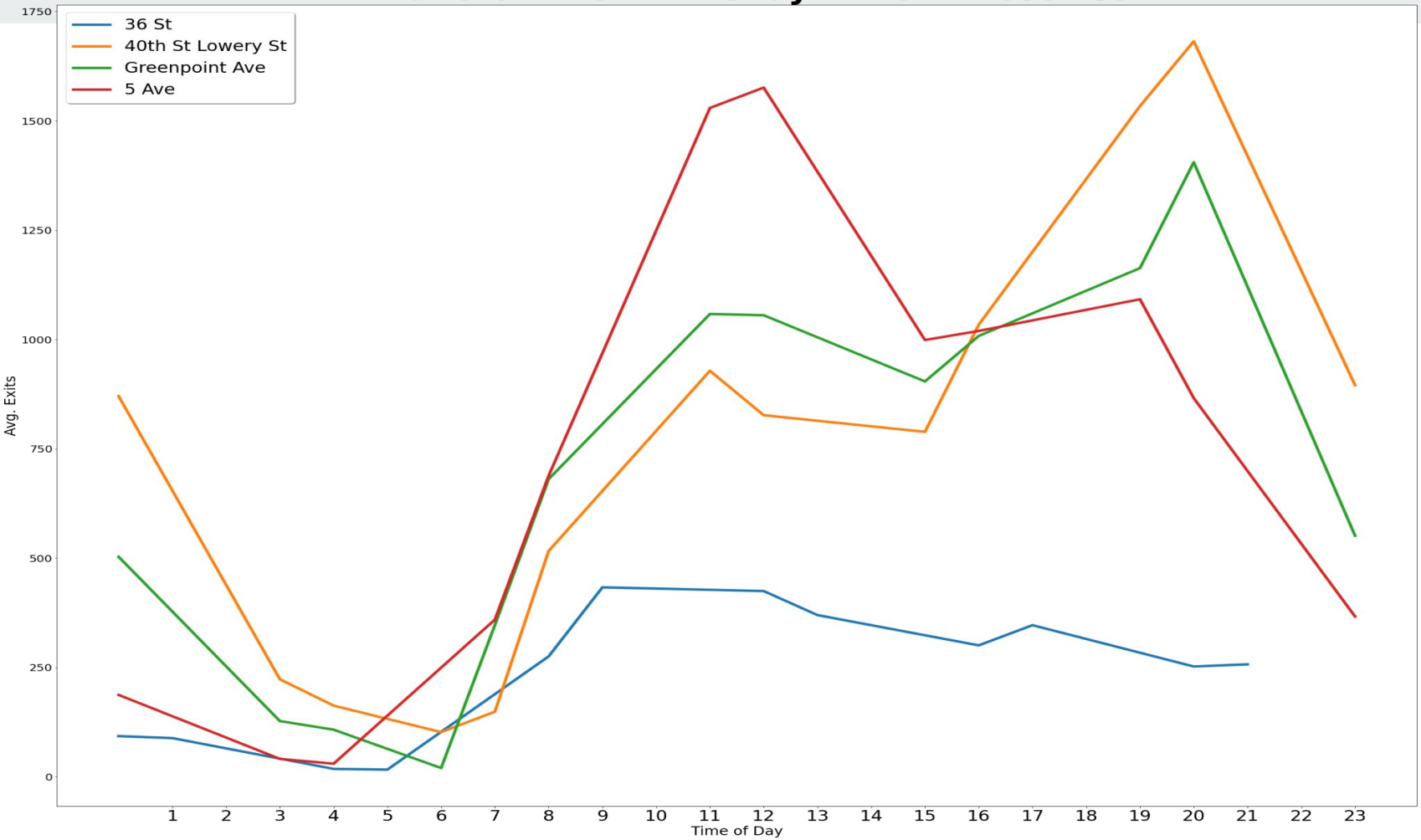
Week Seasonality by Quartile



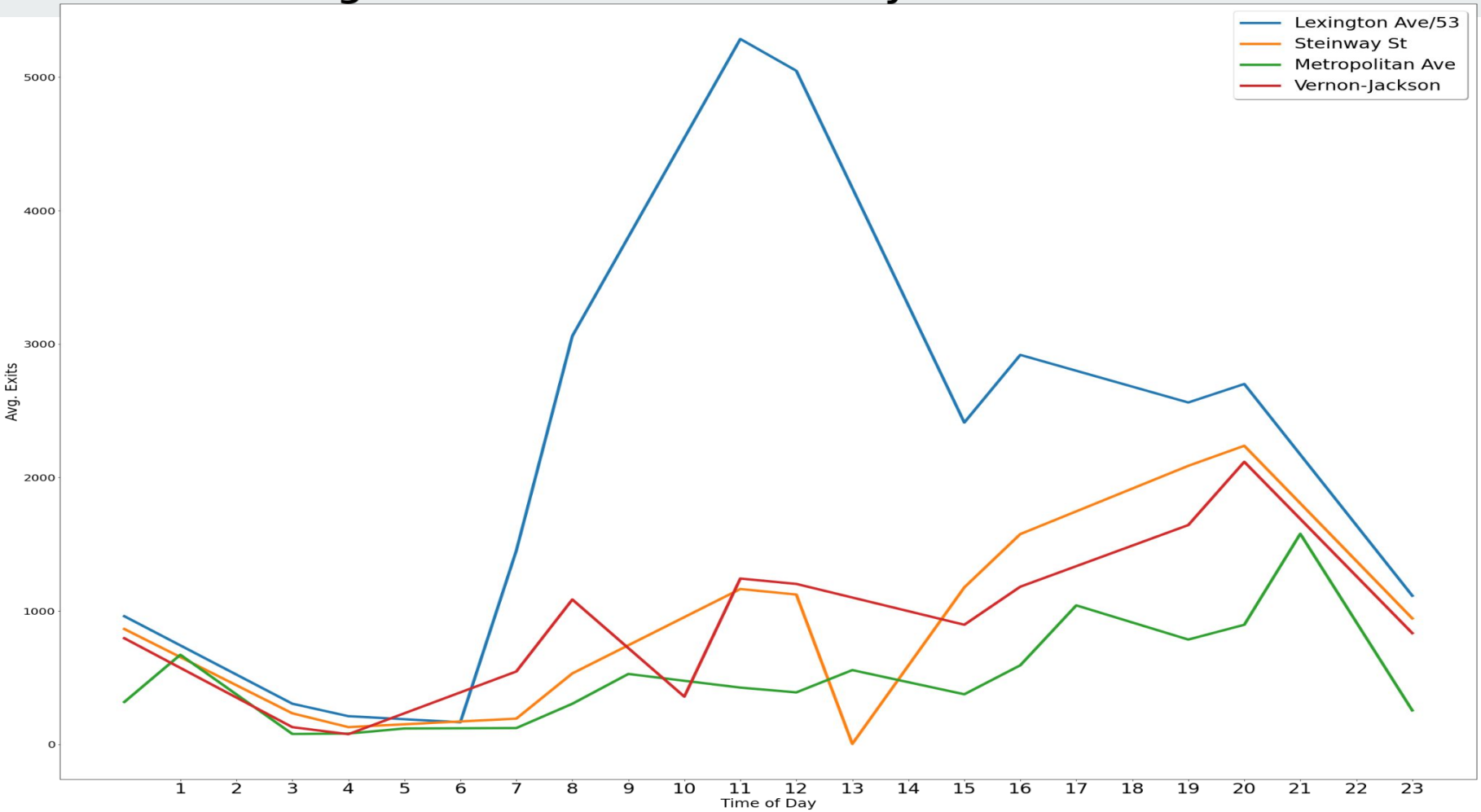

Court Square Week Seasonality
Exits
Entries
Avg. Daily Entries
7000
6000
5000
4000
3000
Monday
Tuesday
Wednesday
Thursday
Friday
Saturday
Sunday
Day of week

Determining Extended Hours

Local Stations: Thursday Exits Timeseries



High Traffic Stations: Thursday Exits Timeseries

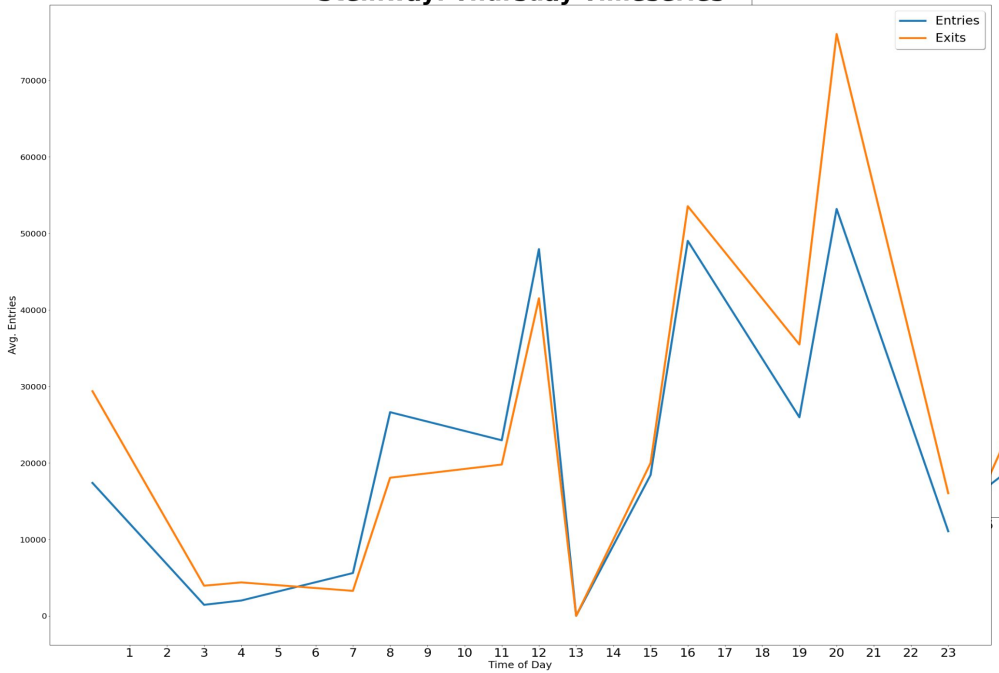


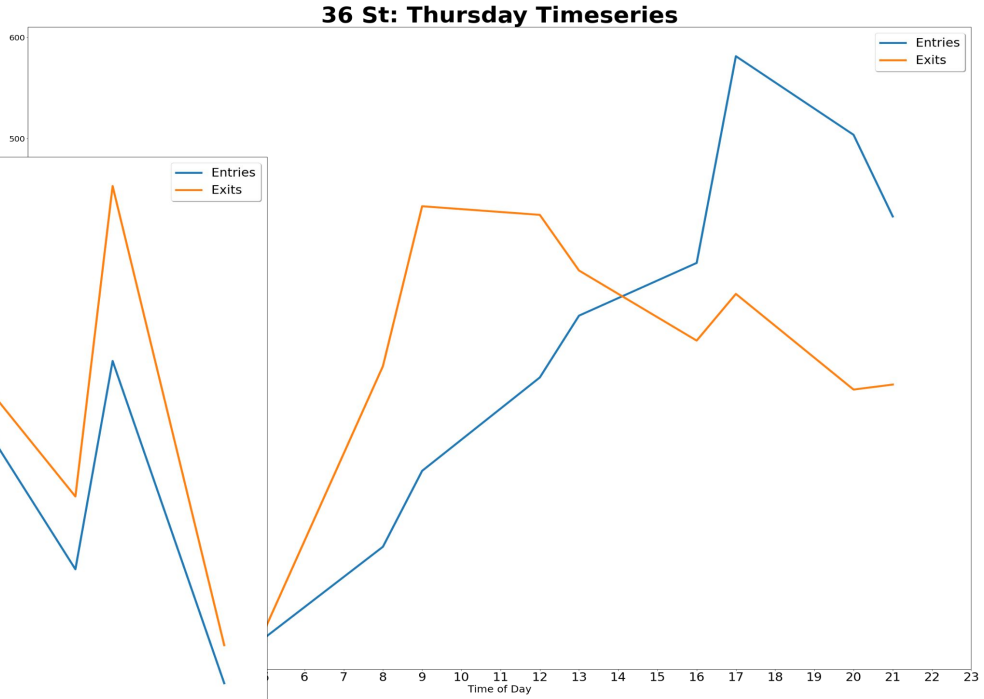
Conclusions

- Thursday is a great day for Late Nights at MOMA PS1
- Exit Traffic peaks around 8 pm for local stations and 10 pm at high-traffic stations
- Late Nights at MOMA PS1 should launch on Thursdays, offering discounted admission until 10.
    - The museum may also consider extending hours on Friday and shortening hours on Saturday and Sunday

Future Work

- Hone in on “real” commuter exits.
  - Pay attention to stations where average exits exceeds entries in the evening.
- Look at weekend traffic patterns
  - Recalibrate weekend hours


Steinway: Thursday Timeseries
Entries
Exits
Avg. Entries
70000
60000
50000
40000
30000
20000
10000
0
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23
Time of Day


36 St: Thursday Timeseries
Entries
Exits
600
500
6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23
Time of Day